

Understanding Plastic Types & Resin Identification Codes

A Comprehensive Guide to Properties, Uses, and Safety

What are Resin Identification Codes (RIC)?

The Global Standard for Sorting and Recycling Plastics

Origin and Purpose

Introduced in 1988 by the Society of the Plastics Industry (SPI) to provide a uniform system for identifying plastic resins in manufacturing.

The Symbol

A number (1-7) inside a chasing-arrows triangle, typically found on the bottom of containers. It helps consumers and facilities identify the base material.

Common Misconception

The presence of the symbol does NOT guarantee recyclability; it only identifies the resin. Local facilities determine what can actually be processed.

Importance

Proper identification is critical for waste management facilities to prevent cross-contamination in the recycling stream.



RIC 1-7

Type 1: PET (Polyethylene Terephthalate)

The Beverage Bottle Standard

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Common Uses

- Water and soda bottles
- Mouthwash and salad dressing containers
- Peanut butter and jam jars
- Polyester fibers for clothing and carpets



Properties & Safety

- ✓ Lightweight, clear, and extremely strong
- ✓ Excellent barrier against oxygen and moisture
- ✓ Highly recyclable and widely accepted



Safety Note

Safe for single use. Avoid high heat (like a hot car) as it may leach antimony trioxide, a potential carcinogen.

Type 2: HDPE (High-Density Polyethylene)

The Durable and Safe Choice

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Properties

- ✓ High strength-to-density ratio
- ✓ Excellent chemical and impact resistance
- ✓ Opaque and more rigid than LDPE
- ✓ Most widely recycled plastic worldwide

Common Uses

- Milk jugs and juice bottles
- Detergent and shampoo containers
- Toys and picnic tables
- Plastic lumber and drainage pipes

Safety Profile

Considered one of the safest plastics with a very low risk of leaching. It is highly stable and resistant to UV rays, making it ideal for long-term storage.

Type 3: PVC (Polyvinyl Chloride)

The "Poison Plastic" - Critical Health & Environmental Risks

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✂ Common Uses

- Plumbing pipes and fittings
- Vinyl flooring and siding
- Medical tubing and blood bags
- Shower curtains and inflatable toys
- Some food wraps and detergent bottles

☠ Health & Safety Risks

- ⚠ Contains **phthalates** which interfere with hormonal development.
- ⚠ May leach heavy metals like **lead** or **cadmium** over time.
- ⚠ Production and incineration released **dioxins**, highly toxic carcinogens.

♻ Recycling Challenges

PVC is **rarely accepted** by curbside programs. It is considered a major contaminant in other recycling streams because its high chlorine content can damage recycling equipment and degrade the quality of other recycled plastics.

Type 4: LDPE (Low-Density Polyethylene)

Flexible and Versatile

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Properties

- ✓ Thinner and more flexible than HDPE
- ✓ Good transparency and moisture resistance
- ✓ Tough and resistant to impact

Common Applications

- Grocery, bread, and frozen food bags
- Squeezable bottles (honey, mustard)
- Plastic wraps and dry cleaning bags
- Coatings for paper milk cartons

Recycling & Safety

Considered safe for food contact. While often not accepted in curbside bins, many retailers provide specialized drop-off points for film plastic recycling.

Type 5: PP (Polypropylene)

The Heat-Resistant Workhorse

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Common Uses

- Yogurt and margarine containers
- Syrup and ketchup bottles
- Medicine bottles and laboratory equipment
- Microwave-safe food containers
- Disposable diaper and sanitary pad liners

Properties & Safety

- ✓ High melting point (approx. 160°C / 320°F)
- ✓ Lightweight, tough, and heat-resistant
- ✓ Excellent resistance to fats and most chemicals
- ✓ Increasingly accepted in curbside recycling

Safety Choice

Considered one of the safest plastics for food and drink. It is the preferred choice for hot-fill liquids and microwave use due to its thermal stability.

Type 6: PS (Polystyrene)

The Lightweight Insulator with Heavy Risks

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Common Uses

- ✓ Disposable coffee cups and lids
- ✓ "Styrofoam" food trays and containers
- ✓ Egg cartons and disposable cutlery
- ✓ Packing peanuts and protective packaging

Health & Environmental Risks

- ❗ **Chemical Leaching:** Can leach styrene, a suspected carcinogen and neurotoxin, especially when heated or in contact with fatty foods.
- ⇒ **Microplastic Pollution:** Structurally weak and brittle; easily breaks into tiny fragments that persist in the environment for centuries.
- 🗑️ **Recycling Barrier:** Extremely low recycling rates due to low density and high transport costs; most curbside programs do not accept it.

Consumer Advice

Avoid #6 plastic for hot food and drinks. It is one of the most problematic plastics for both human health and marine ecosystems.

Type 7: Others (BPA & Bioplastics)

The Miscellaneous "Catch-All" Category

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≡ What's Included?

- ✓ Polycarbonate (PC)
- ✓ Acrylic and Nylon
- ✓ Fiberglass and specialized resins
- ✓ Modern Bioplastics (e.g., PLA)
- ✓ Multi-layer or mixed plastics

⚠ Key Considerations

- 🧪 **BPA Risks:** Polycarbonate often contains Bisphenol A (BPA), an endocrine disruptor linked to hormonal and developmental issues.
- 🌱 **Bioplastics:** Materials like PLA are made from corn starch. While "compostable," they often require industrial facilities to break down.
- 🚫 **Recycling:** Extremely difficult to recycle as it contains a mix of unknown resins. Most curbside programs do not accept Type 7.

📌 Consumer Tip

Use Type 7 with caution for food and drink unless labeled "BPA-Free." For bioplastics, check if they are certified for home composting or industrial only.

Plastic Safety and Recyclability Comparison

Quick Reference for Consumer Choices

Plastic Type	Safety Rating	Recyclability	Heat Resistance
1 - PET	⚠ Moderate (Single use)	✓ High	Low
2 - HDPE	✓ High	✓ High	Moderate
3 - PVC	✗ Low (Avoid)	✗ Very Low	Moderate
4 - LDPE	✓ High	ℹ Moderate	Low
5 - PP	✓ High	ℹ Moderate	High
6 - PS	✗ Low (Avoid)	✗ Low	Low
7 - Other	❓ Varies (Caution)	✗ Very Low	Varies

Environmental Impact of Plastics

Beyond the Bin: A Global Challenge

Persistence

Most plastics take 400 to 1,000 years to decompose in landfills. They don't biodegrade; they only break down into smaller pieces.

Microplastics

Physical breakdown creates micro-particles found in soil, water, and human blood, with unknown long-term health effects.

Marine Pollution

Over 8 million tons of plastic enter the oceans annually, harming marine life through ingestion and entanglement.

Carbon Footprint

Plastic production is energy-intensive and relies heavily on fossil fuel extraction, contributing significantly to global emissions.

The Scale of the Crisis

Since the 1950s, over 8.3 billion metric tons of plastic have been produced. Only 9% has been recycled, while 79% has accumulated in landfills or the natural environment.

Sustainable Best Practices

Reducing Your Plastic Footprint and Ensuring Safe Usage

The "Safe" Three

When using plastic for food or drink, prioritize these types as they have the lowest risk of chemical leaching:

Type 2 (HDPE)

Type 4 (LDPE)

Type 5 (PP)

Refuse and Reuse

Prioritize reusable glass, stainless steel, or bamboo containers. Carry your own shopping bags and water bottles to eliminate the need for single-use plastics.

Avoid Heat Exposure

Never microwave plastics unless explicitly labeled "microwave-safe." Avoid leaving plastic water bottles in hot cars or direct sunlight to prevent chemical migration.

Check Local Guidelines

Recycling rules vary significantly by city. Always verify which RIC codes your local facility actually processes to avoid "wish-cycling" and contamination.

"The most sustainable plastic is the one you choose not to use."